Improving Declustering for Qcify AIR

Reflection

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1. Introduction

This reflection document looks back on the internship project carried out at Qcify. The project focused on investigating possible improvements for almond declustering in the Qcify AIR machine. At the start of the project, there was no declustering segmentation model actively used in production for this purpose. There had been previous experiments, but they did not provide reliable enough results to be used in the machine. More specifically, the goal was to investigate how closely positioned almonds could be separated more efficiently and reliably, while keeping the model fast enough for a real-time industrial environment.

The internship assignment was approached as a research and development project. This means, there was no fixed solution to deliver, but to explore different technical directions, compare them and identify which approaches showed the most potential. During the internship, I worked on dataset preparation, boundary and distance-map generation, training different model configurations, evaluating the results and creating visualisations to understand the model behaviour better.

This document contains two main reflections. The first part is a substantive reflection on the project itself: what was achieved, what value it provides for Qcify, what remains open and what could be done in the future. The second part focuses on personal reflection on what I learned during the internship, which skills I developed and which challenges I faced.

2. Substantive reflection

During the internship I investigated several approaches for improving almond declustering in the Qcify AIR segmentation pipeline. The starting point was not a production segmentation model, but an experimental direction that had been tested before. Earlier declustering experiments showed that separating almonds in the segmentation output was possible, but the results were not reliable enough for production use. One of the main problems was that stronger separation could remove too much useful information from almond regions, which could negatively affect later classification.

The main contribution of the project was extending this setup into a more informative boundary-aware approach. Instead of representing the segmentation target only as background and nut pixels, I worked with a three-class representation: background, nut boundary and nut interior. This gave the model more information about the structure and about the areas where neighbouring nuts should be separated.

Another important part of the project was the preparation of training data. I worked with almond images captured from the Qcify AIR machine and prepared datasets with different input heights and target types. The datasets included boundary masks and, in some experiments, distance maps. Boundary masks were used to indicate the contour area of the objects, while distance maps provided additional information about the internal structure. I also extended the dataset preparation workflow in a Jupyter Notebook, which made the process of generating annotations, merging datasets and creating training data easier and repeatable.

Most of the model experiments were based on ESPNetV2, because it is a lightweight architecture and therefore suitable for a real-time industrial environment. Different configurations were trained and evaluated, including boundary-only setups, distance maps and boundary masks setups, different input heights and different loss functions. In addition, I also tested penalty terms that were meant to reduce specific failure cases, such as predicting almond interior on boundary pixels or predicting boundaries inside almond interiors.

Besides ESPNetV2 experiments, I also implemented and tested an FBSNet-based model. This was not the main direction of the project, but it was useful as an additional comparison. The FBSNet results showed that alternative lightweight architectures could be interesting for future research, although more experiments would be needed before drawing any conclusions about this model.

The evaluation was done using both quantitative metrics and visual inspection. This was important because the quality of declustering cannot be judged only by standard segmentation metrics. A model can score high in a metric such as IoU, while still merging neighbouring objects into one connected component. Because of that, I also compared object-level metrics such as component count error, merge rate and split rate. In addition, I inspected the model with generated visualisations of predicted masks to understand issues such as holes inside masks, merged nuts, over-segmentation and incorrect boundary placement.

The project did not result in a production-ready declustering model, but it did provide Qcify with a structured comparison of different technical directions. The results showed which configurations are worth further investigation and which trade-offs are important. In particular, the results showed that models with images of higher input size produce cleaner pixel-level masks, while the models trained with smaller images are more interesting from a runtime and object-counting perspective.

For Qcify, the value of this work is mainly in the research outcome. The project results give a clearer understanding of how boundary-aware segmentation behaves on almond data and which model configurations are the most promising. It also provides trained models, evaluation results, visual examples and a more automated workflow for preparing datasets and evaluating future models.

The project still needs further work before it could be considered for production use. The most important next step would be to test the models that are the strongest candidates on a bigger dataset and over longer production-like runs.

My advice for future work is to continue with the most promising ESPNetV2 configurations first, because they are the closest to the runtime requirements of the machine. At the same time, FBSNet or other lightweight architectures could be investigated further, but only with more training runs and a fair comparison setup. It could also be useful to test different boundary thickness, since all the current experiments used the same fixed size. Finally, post-processing methods such as small-object removal or connected-component filtering could be explored to improve the final object separation.

3. Personal reflection

This internship was a valuable experience for me because it showed what it is like to work on a real computer vision problem in an industrial environment. The project was different from many school assignments, because there was no single correct solution from the beginning. It was a research-oriented task, so I had to try different approaches, compare results and accept that not every experiment will lead to an improvement.

One of the most important things I learned was that evaluating a model is not as simple as only looking at a couple of metrics. In the beginning, I mostly focused on IoU, since it is a common metric for segmentation. During the project, I quickly realised that this is not enough. A model can have a high IoU and still merge objects together, while a model with a lower IoU result can be more useful in practice. This helped me understand that evaluation has to be connected to the real goal of the project.

I also learned a lot about data preparation. Before the internship, I did not fully realise how much work is needed before a model can be properly trained. I worked with annotations, automatic mask generation, boundary masks, distance maps, slicing images to produce smaller input sizes and dataset merging. This made me understand that the quality and structure of the data strongly influence the final model behaviour. I also learned that automation is useful, but it still requires manual checking, especially when annotations are generated automatically.

Another challenge I faced was implementing an additional FBSNet architecture. Unlike ESPNetv2, which was already available in the training pipeline, FBSNet had to be implemented and adapted manually. It was difficult for me because there was limited practical information about this architecture that was available online, especially in the form that could be directly reused. I had to rely on the available paper, the GitHub repository, the information and feedback I received from my mentor, my own understanding of encoder-decoder segmentation networks and prompt-engineering. This helped me understand how research-based architectures can be translated into working code and adapted to an existing project.

Another important learning point was working with an existing company codebase. In the beginning, it was difficult for me to understand how all parts of the pipeline were connected. Over time, I became more confident in reading existing code, modifying it and adding my own changes.

I also faced the challenge of keeping track of many similar experiments. Different runs used different datasets, input sizes, losses and penalty terms. Sometimes the differences between runs were small, but they still influenced the results. This taught me how important it is to document experiments properly. Without proper notes, it becomes very difficult to later understand why one model behaved differently from another.

I also had to deal with results that were not easy to interpret. Some models looked better visually but had worse object-counting metrics and vice versa. This was sometimes frustrating but helped me become more systematic. Instead of immediately deciding which model was the best, I learned to compare them from multiple perspectives.

This internship also helped me improve my communication. I had to explain technical results, discuss trade-offs and understand feedback from people with more experience. This was important since the project was not only about training models, but also about understanding what would actually be useful for the company.

I would also like to thank my mentor, Dries Hendrikx, for his guidance, feedback and support during the internship. His explanations and suggestions helped me better understand the technical challenges of the project and gave me more confidence when working with complex implementation and evaluation tasks.

Overall, this internship helped me grow both technically and personally. I learned more about segmentation, model training, dataset preparation, evaluation metrics and real-time constraints. I also became more comfortable working independently and analysing results critically. Even though the project did not end with a fully production-ready solution, it gave useful insights for Qcify and gave me valuable experience in applying computer vision to a real industrial problem.